

Priyansha

Mechanical Engineer (Robotics & Automation)

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[LinkedIn](#) | [Personal Website](#) | [Design Portfolio](#)

Education

Malaviya National Institute of Technology Jaipur

2022 – 2026

B.Tech Mechanical Engineering (Honours in Robotics and Automation)

CGPA: 8.06/10

Navrachana Higher Secondary School (CBSE), Vadodara

2014 – 2021

Class 12: 92.8% | Class 10: 95.5%

Achievements: IIT Bombay Techfest Finalist, TCS IT Wiz Runner-up

Work Experience

Mechanical Intern

May 2025 – July 2025

Reude Technologies | Remote

- Conducted feasibility analyses for an automated drone delivery station, optimising manufacturability and robustness.

Robotics Intern

June 2024 – Aug 2024

TIH iHub Drishti | Jodhpur

- Benchmarked 4 vSLAM methods (Monocular, Mono-Inertial, Stereo, Stereo-Inertial) using ROS2
- Developed simulations in ROS and Gazebo; worked with embodied AI for autonomous path planning of the Clearpath Husky UGV.

Intern

May 2023 – Sep 2023

Insights Autotech Consultants | Remote

- Conducted market research and mechanical design tasks related to EV systems.
- Conducted market research and supported the development of sales strategies.

Skills

- Mechanical & CAD: SolidWorks, Fusion360, AutoCAD, Ansys, ABAQUS
- Robotics & Control: ROS2, Gazebo, CoppeliaSim, Arduino, Control Systems, MATLAB
- Programming: Python, MATLAB, GNU Octave, C
- Others: Operations Research, Technical Documentation

Projects

Design and Control of Quadruped Robot

Aug 2025 – Present

- Design and prototyping of a quadruped spider robot for coordinated locomotion.
- Developed a 3D-printable rotating LiDAR mount in SolidWorks.

Robot Arm (Prototype)

Jan 2025 – Present

- Designed a 3D-printed robot arm using servo motors and ESP32 based control.
- Implemented Arduino, MATLAB and inverse kinematics for movement control.

Compact High-Torque Gearbox

April 2025

- Designed a 20:1 gearbox for robotics applications with a torque output of 50–100 Nm within size / weight limits.
- Conducted FEA and introduced a self-locking wear-resistant mechanism.

E-Yantra Robotics Competition

Sep 2023 – Jan 2024

- Simulated a self-balancing bike robot using rotary inverted pendulum dynamics.
- Designed and tuned a PID controller in CoppeliaSim, integrated with Arduino & Python.

Achievements and Interests

- Cleared GATE 2025
- Trained in Bharatnatyam (6 years) and Hindustani Classical Music (5 years)
- Passionate about ecological restoration, sustainability, and creative expression (poetry, writing)